THERMODYNAMICS

1. Consider the following two statements
(A) If heat is added to a system, its temperature must increase.
(B) If positive work is done by a system in a thermodynamic process, its volume must increase.
(1) Both A & B are correct
(2) A is correct but B is wrong
(3) B is correct but A is wrong
(4) Both A & B are wrong

2. A gas is contained in a metallic cylinder fitted with a piston. The piston is suddenly moved in to compress the gas and is maintained at this position. As time passes the pressure of the gas in the cylinder :-
(1) Increases
(2) Decreases
(3) Remains constant
(4) Depends on the nature of gas

3. Which of the following can be called as thermodynamic property.
(A) $\int PdV$ (B) $\int VdP$ (C) $\int PdV + \int VdP$
(1) Both A & B
(2) Both B & C
(3) All three
(4) Only C

4. For any reversible process, the change in the entropy of the system and surroundings is
(1) 0 (2) 1
(3) ∞ (4) None of these

5. Which of the following is correct according to Claussius statement of second law of thermodynamics.
(1) It is possible to transfer heat from a low temp. reservoir to a high temp. reservoir.
(2) No heat engine can have 100% efficiency.
(3) No process is possible whose sole result is the transfer of heat from a colder object to a hotter object
(4) None of these

6. In T-S diagram, ratio of slope of an isobaric curve and an isochoric curve is.
(1) γ (2) $\frac{1}{\gamma}$ (3) γ^2 (4) Zero

7. Consider four heat reservoirs A, B, C, D. An engine working between A and C has an efficiency which is mean of efficiencies of same engine working between A, B and A, D. Then absolute temperture of C will be : (A is source)
(1) Geometric mean of temperature of A & D
(2) Arithmetic mean of temperature of B & D
(3) Harmonic mean of temperature of A & D
(4) Sum of temperatures of A and D.

8. Which of the following statements is correct for any thermodynamic system
(1) internal energy changes in all processes.
(2) internal energy and entropy are state functions.
(3) change in entropy can never be zero.
(4) work done in an adiabatic process is always zero.

9. Which of the following statement is incorrect.
(1) All reversible cycles have same efficiency.
(2) Reversible cycle has more efficiency than an irreversible one.
(3) Carnot cycle is a reversible one.
(4) Carnot cycle has the maximum efficiency in all cycles.

10. Molar specific heat capacity of substance does not depend on-
(1) Nature of the substance
(2) Temperature of the substance
(3) Amount of the substance
(4) Condition under which heat is supplied

11. Which of the following is meaningful statement
(1) A gas in a given state has a certain amount of heat
(2) A gas in a given state has a certain amount of work
(3) A gas in a given state has a certain amount of internal energy
(4) all statements are meaningless

12. One calorie is defined to be the amount of heat required to raise the temperature of 1g of water from ..... at 1 atm.

(1) 0°C to 1°C (2) 10°C to 11°C

(3) 4°C to 5°C (4) 14.5°C to 15.5°C

13. Which of the following statement is not correct-

(1) The efficiency of the carnot heat engine depends only on the temperatures of the source and sink between which the engine works.

(2) The efficiency of a heat engine can never be unity.

(3) For a refrigerator, the co-efficient of performance can never be infinite

(4) None of these

14. The pressure P and volume V of an ideal gas both increase in a process-

(i) Such a process is not possible

(ii) The work done by the system is positive

(iii) The temperature of the system must increase.

(iv) Heat supplied to the gas is equal to the change in internal energy

choose correct statements :-

(1) (i), (ii) (2) (ii), (iii)

(3) (iii), (iv) (4) (i), (iv)

15. Which of the following statement is correct-

(1) Extensive variables are internal energy (U), Volume (V) and Pressure (P).

(2) Intensive variables are pressure (P), Temperature (T) and total mass (M)

(3) Extensive variables are volume (V), total mass (M) and internal energy (U)

(4) Intensive variables are temperature, density(ρ) and volume (V)

6. In case of water from 0°C to 4°C

(i) Volume decreases and density of water is maximum at 4°C

(ii) Δ work will be negative, since volume decreases

(iii) $C_p > C_v$

(iv) $C_p < C_v$

Choose correct statements :-

(1) (i), (ii) (2) (ii), (iii)

(3) (iii), (i) (4) (i), (ii) & (iv)

17. Which of the following statement is correct.

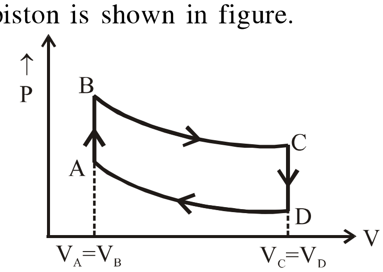
(1) If heat is added to a system, its temperature must increase.

(2) If positive work is done by a system in a thermodynamic process, its volume must increase.

(3) In free expansion, heat is necessarily absorbed by the system.

(4) In a process, the initial pressure and volume are equal to the final pressure and volume then the net work done by the system in the process must be zero.

18. A cycle followed by an engine [made of one mole of perfect gas $\left(\gamma = \frac{5}{3}\right)$ in a cylinder with a piston is shown in figure.



A to B : Volume constant

B to C : Adiabatic

C to D : Volume constant

D to A : Adiabatic

and $V_C = V_D = 2V_A = 2V_B$

which of the following statement is wrong -

(1) In AB part of the cycle heat is supplied to the engine from outside

(2) In CD part of the cycle heat is being given to the surrounding by the engine

(3) In DA part of the cycle internal energy of the system increases

(4) In BC part of the cycle internal energy of the system increases

19. Heat is added to an ideal gas and the gas expands. In such a process the temperature-

(1) must increase (2) will remain same

(3) must decrease (4) can not say

20. Under isobaric condition if the temperature of a room increases then-

(1) Total KE of the molecules increases.

(2) Total KE of the molecules decreases.

(3) Total KE of the molecules remain same.

(4) The density of air increases.

21. An ideal gas contained in a cylinder by a frictionless piston is allowed to expand such that temperature remains constant then work done by the gas-

(1) Zero
(2) Positive
(3) Negative
(4) May increase or decrease

22. During the melting of a slab of ice at 273K at atomospheric pressure-

(1) Work done by ice plus water system on atmosphere is positive
(2) Work done on ice plus water system by atmosphere is positive
(3) Internal energy of ice plus water system increases
(4) Both (2) and (3)

23. For a given process on an ideal gas $dW = 0$ & $dQ < 0$ then for the gas-

(1) Temp will decrease
(2) Vol. will increase
(3) Pressure will remain constant
(4) Temp will increase

24. A glass of water is stirred and then allowed to stand until the water stops moving. Then internal energy of water -

(1) remains constant
(2) will increase
(3) will decrease
(4) nothing can be said

25. If an ideal gas is compressed suddenly then-

(1) It's temperature will increase
(2) It's temperature will decrease
(3) Temperature remains constant
(4) None of the above

26. If a heat engine floating on the ocean extract heat from water and convert into mechanical work. Is it possible?

(1) Yes, from 1^{st} law of thermodynamic
(2) No, from 1^{st} law of thermodynamic
(3) Yes, from 2^{nd} law of thermodynamic
(4) Yes, from both 1^{st} and 2^{nd} law of thermodynamic

27. For a spontaneous process of the system, where S is entropy.

(1) $\Delta S > 0$ (2) $\Delta S = 0$
(3) $\Delta S < 0$ (4) None of the above

28. A piston is slowly pushed into a metal cylinder isothermally containing an ideal gas then incorrect statement is-

(1) Pressure of the gas increases.
(2) Average speed of the gas molecules increases.
(3) The number of molecules per unit volume increases
(4) None of these

29. Two vessel of the same volume contain the same gas at same temperature. If pressure in the vessel are in ratio of 1 : 2 then-

(1) Ratio of v_{rms} is 1 : 2
(2) Ratio of kinetic energy is 1 : 1
(3) Ratio of average velocity is 1 : 2
(4) Ratio of number of molecules is 1 : 2